Mutual Fund Analytics Platform

End-to-End Data Engineering, Performance Analytics and Business Intelligence Dashboard

Capstone Project Report

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Executive Summary

- The Indian mutual fund industry has experienced rapid growth in assets under management, investor participation, and systematic investment plans (SIPs). As the volume of financial data continues to increase, extracting meaningful insights from this information has become increasingly important for fund managers, analysts, and investors.

- This project presents an end-to-end Mutual Fund Analytics Platform that integrates data engineering, financial performance analysis, advanced risk analytics, and interactive business intelligence. Multiple datasets containing mutual fund information, NAV history, investor transactions, benchmark indices, portfolio holdings, and industry statistics were processed through a structured ETL pipeline before being stored in a SQLite database for analysis.

- The project computes key financial metrics including Daily Returns, CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Tracking Error, and a composite Fund Score. Advanced analytical techniques such as Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration using the Herfindahl-Hirschman Index (HHI), and a rule-based Mutual Fund Recommendation System were also implemented.

- Finally, an interactive Power BI dashboard was developed to present industry trends, fund performance, investor analytics, and SIP market insights through interactive visualizations and drill-through capabilities. The completed platform demonstrates how financial data can be transformed into meaningful insights that support informed investment decisions.

Introduction

Mutual funds are among the most widely used investment instruments because they provide diversification, professional fund management, and accessibility to investors with varying financial goals and risk appetites. As investment portfolios become larger and financial markets more dynamic, analyzing fund performance using quantitative techniques has become increasingly important.

Traditional analysis often focuses on returns alone, which does not provide a complete understanding of investment quality. Modern portfolio evaluation considers both return and risk using metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown. These indicators allow investors to evaluate whether returns adequately compensate for the risks undertaken.

This project was developed to create a comprehensive analytics platform capable of processing mutual fund datasets, generating performance metrics, analyzing investor behavior, and presenting results through interactive dashboards. By combining data engineering, statistical analysis, and business intelligence techniques, the project provides a complete workflow from raw financial data to actionable insights.

Objectives

The primary objectives of this project are:

- Build an end-to-end ETL pipeline for processing mutual fund datasets.
- Clean and standardize financial and investor data.
- Store processed data within a SQLite database.
- Perform exploratory data analysis to identify industry trends.
- Compute key financial performance metrics for mutual funds.
- Implement advanced analytical techniques for risk assessment and investor behavior analysis.
- Develop a rule-based recommendation system for mutual fund selection.
- Design an interactive Power BI dashboard for data visualization and decision support.
- Present the complete workflow through professional documentation and reporting.

Dataset Overview

The project uses multiple datasets representing different aspects of the Indian mutual fund ecosystem. Together, these datasets provide information on fund characteristics, historical NAV values, investor transactions, benchmark performance, portfolio composition, and overall industry statistics. After ingestion, each dataset was cleaned, standardized, and integrated into the analytics pipeline.

Table 4.1: Datasets Used

Dataset	Description
Fund Master	Contains scheme metadata including AMFI code, fund house, category, and plan information.
NAV History	Historical daily Net Asset Value (NAV) records used for return and risk calculations.
AUM by Fund House	Monthly Assets Under Management (AUM) statistics for different asset management companies.
Monthly SIP Inflows	Industry-wide monthly SIP inflow data used for trend analysis.
Category Inflows	Net inflows across mutual fund categories such as Equity, Debt, Hybrid, and Liquid funds.
Industry Folio Count	Monthly folio statistics indicating investor participation across the industry.
Scheme Performance	Fund-level performance indicators including returns, volatility, risk grade, expense ratio, and ratings.
Investor Transactions	Synthetic investor transaction records containing SIPs, redemptions, demographics, and investment behavior.

Dataset	Description
Portfolio Holdings	Portfolio composition of mutual funds including stock weights and sector allocations.
Benchmark Indices	Historical benchmark index values used for Alpha, Beta, and Tracking Error calculations.

Data Characteristics

The datasets cover multiple years of mutual fund activity and contain both time-series and tabular information. Key characteristics include:

- Daily historical NAV values for multiple mutual fund schemes.
- Monthly industry-level investment statistics.
- Investor demographic and transaction information.
- Portfolio holdings and sector allocation.
- Benchmark index data for comparative performance analysis.
- Fund metadata including category, risk grade, and expense ratio.

All datasets were cleaned and standardized before being used for exploratory analysis, performance analytics, advanced analytics, and dashboard development.

ETL Pipeline

An Extract, Transform, and Load (ETL) pipeline was developed to automate the preparation of raw mutual fund datasets for analysis. The pipeline ensures that data from multiple sources is consistently cleaned, transformed, and stored before being used in downstream analytical workflows.

ETL Workflow

```
Raw CSV Files
      |
      ▼
Data Ingestion
      |
      ▼
Data Cleaning & Validation
      |
      ▼
Feature Engineering
      |
      ▼
SQLite Database
      |
      ▼
Performance Analytics
      |
      ▼
Advanced Analytics
      |
      ▼
Power BI Dashboard
```

Extract

The pipeline begins by loading raw datasets from CSV files, including fund information, NAV history, investor transactions, portfolio holdings, benchmark indices, and industry statistics. Additional live NAV data is also retrieved using the MFAPI service to supplement historical records.

Transform

During the transformation stage, several preprocessing operations are performed:

- Removal of duplicate records.
- Handling of missing values and inconsistent entries.
- Standardization of column names and data types.
- Date parsing and formatting.
- Forward filling of missing NAV values where appropriate.
- Generation of cleaned datasets for downstream analytics.
- Creation of derived metrics and analytical features.

Load

The cleaned datasets are loaded into a SQLite database using a predefined relational schema. SQL scripts were developed to create the required tables and support efficient querying throughout the project. The processed datasets also serve as inputs for the performance analytics notebook, advanced analytics notebook, and Power BI dashboard.

Technologies Used

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Jupyter Notebook
- Power BI

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the mutual fund datasets before conducting performance and risk analytics. The analysis focused on identifying long-term industry trends, investor behavior, portfolio composition, and benchmark performance. Visualizations were generated using Python libraries such as Matplotlib and Seaborn to support data interpretation.

6.1 Net Asset Value (NAV) Trend

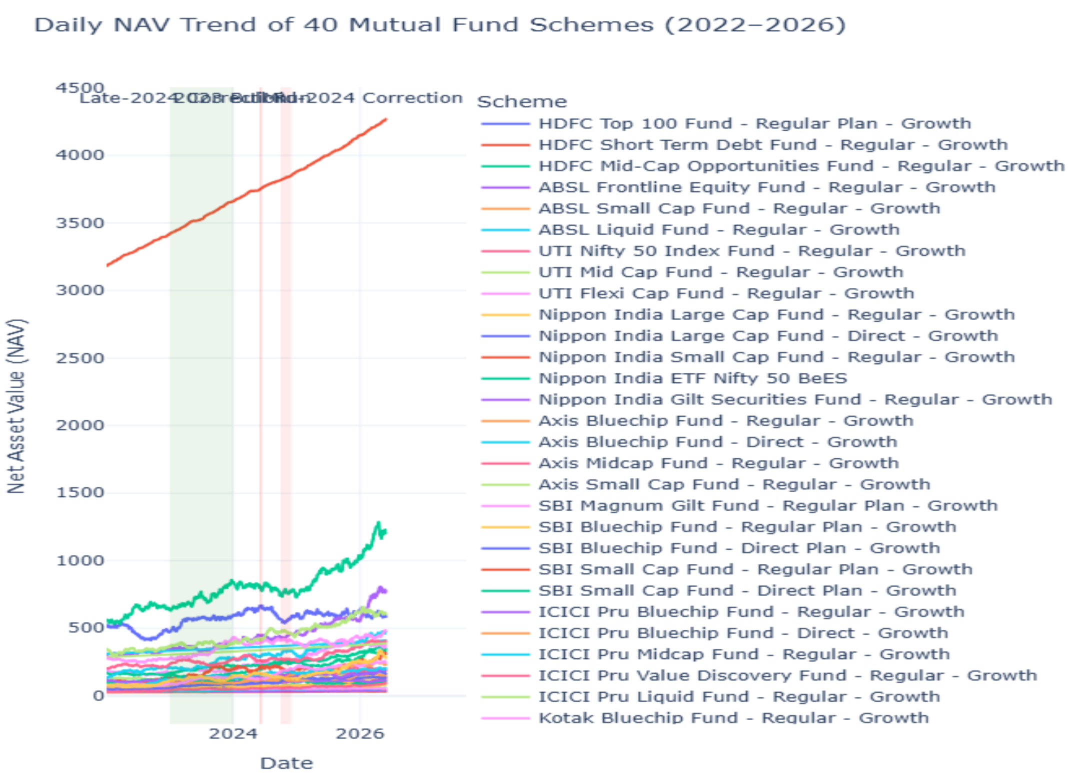


Figure 6.1: Historical NAV Trend

Description

The historical NAV trend illustrates the growth in fund values over the observation period. Most schemes exhibited a steady upward trajectory despite short-term market fluctuations, reflecting long-term capital appreciation and the benefits of disciplined investing.

6.2 Assets Under Management (AUM) Growth

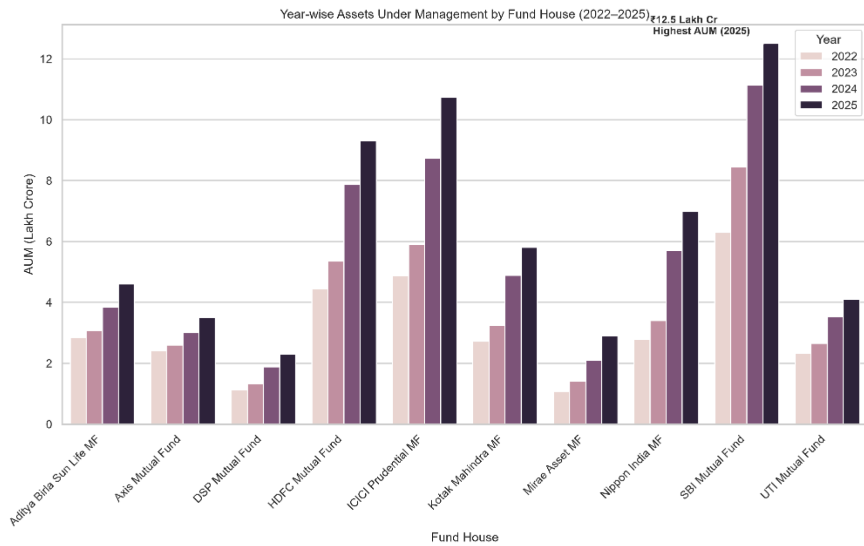


Figure 6.2: Industry AUM Growth

Description

The AUM trend indicates continuous expansion of the mutual fund industry, demonstrating increasing investor participation and confidence. The sustained rise in managed assets highlights the growing importance of mutual funds within the Indian financial market.

6.3 Monthly SIP Trend

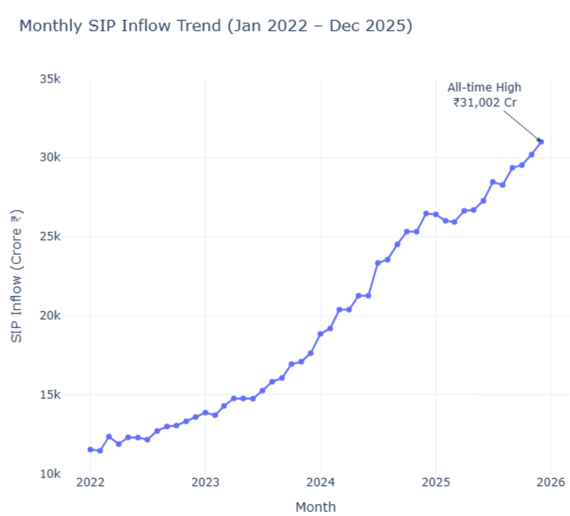


Figure 6.3: Monthly SIP Inflows

Description

Monthly SIP inflows displayed a consistent upward trend throughout the study period. This reflects increasing adoption of systematic investment plans as a preferred long-term investment strategy among retail investors.

6.4 Category Inflow Analysis

Figure 6.4: Category-wise Inflows

Description

The heatmap highlights variations in fund inflows across different mutual fund categories. Equity-oriented categories generally attracted stronger investor interest, while debt and liquid funds experienced comparatively stable investment patterns.

6.5 Investor Demographics

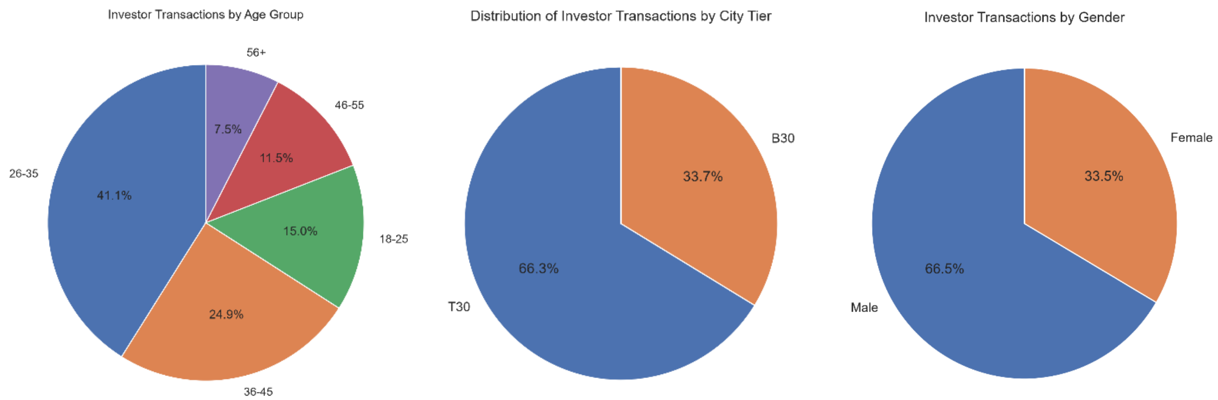


Figure 6.5: Investor Demographic Distribution

Description

Investor demographics reveal participation across multiple age groups, genders, and city tiers. These distributions provide valuable context for understanding investment behavior and support subsequent investor analytics.

6.6 Industry Folio Growth

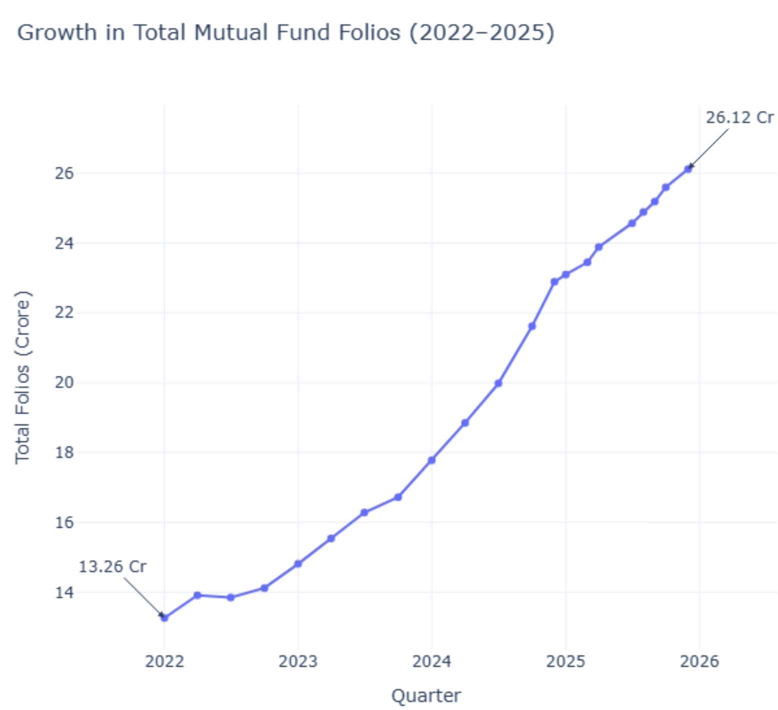


Figure 6.6: Growth in Mutual Fund Folios

Description

The increasing number of mutual fund folios indicates expanding retail investor participation. This trend aligns closely with the observed increase in SIP inflows and industry AUM.

6.7 Portfolio Sector Allocation

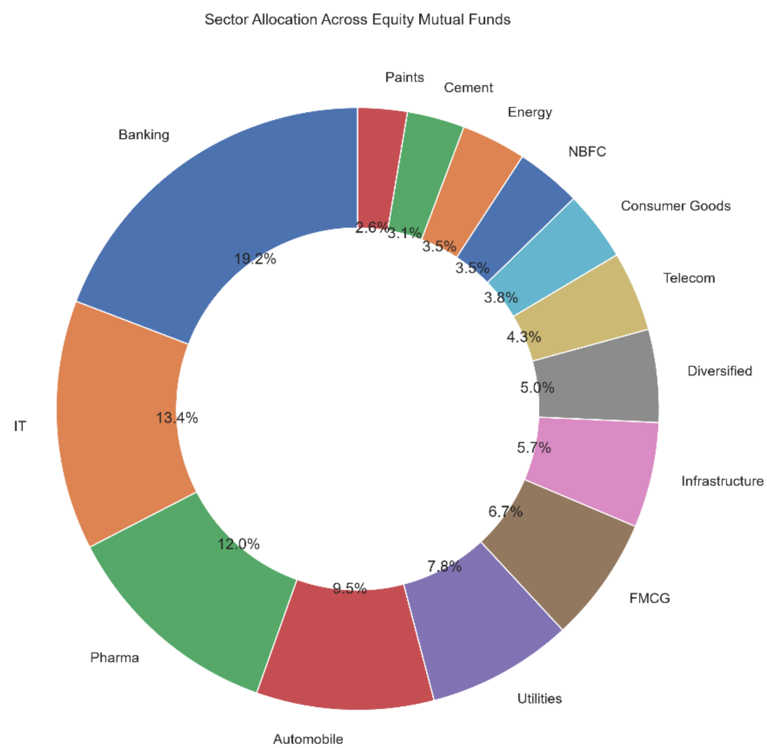


Figure 6.7: Sector Allocation Across Portfolios

Description

Sector allocation analysis illustrates how mutual fund investments are distributed across different industries. Diversification across sectors helps reduce concentration risk while providing exposure to multiple segments of the economy.

6.8 Correlation Analysis

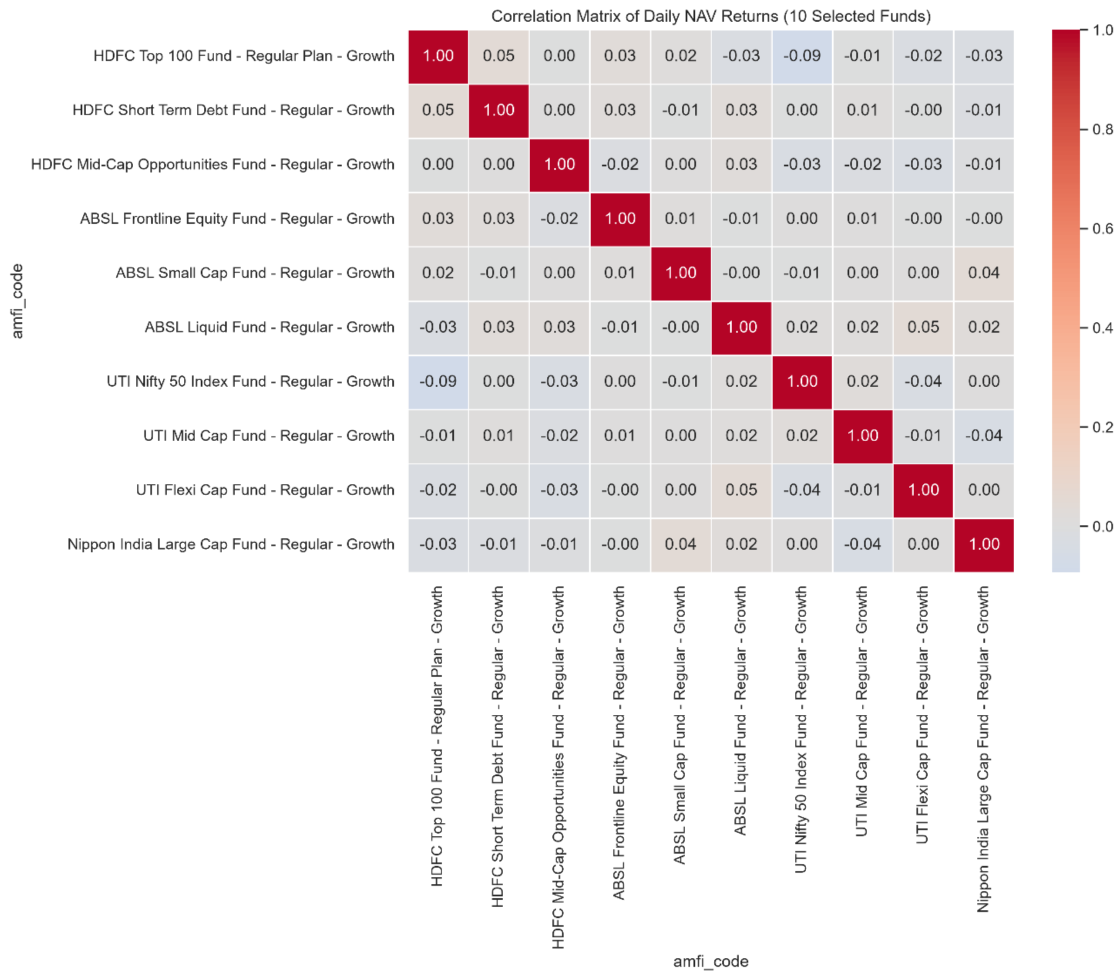


Figure 6.8: Correlation Heatmap of Fund Returns

Description

The correlation heatmap measures the degree of similarity in returns among selected mutual funds. Highly correlated funds tend to move together under similar market conditions, whereas lower correlations indicate better diversification opportunities.

6.9 Benchmark Performance

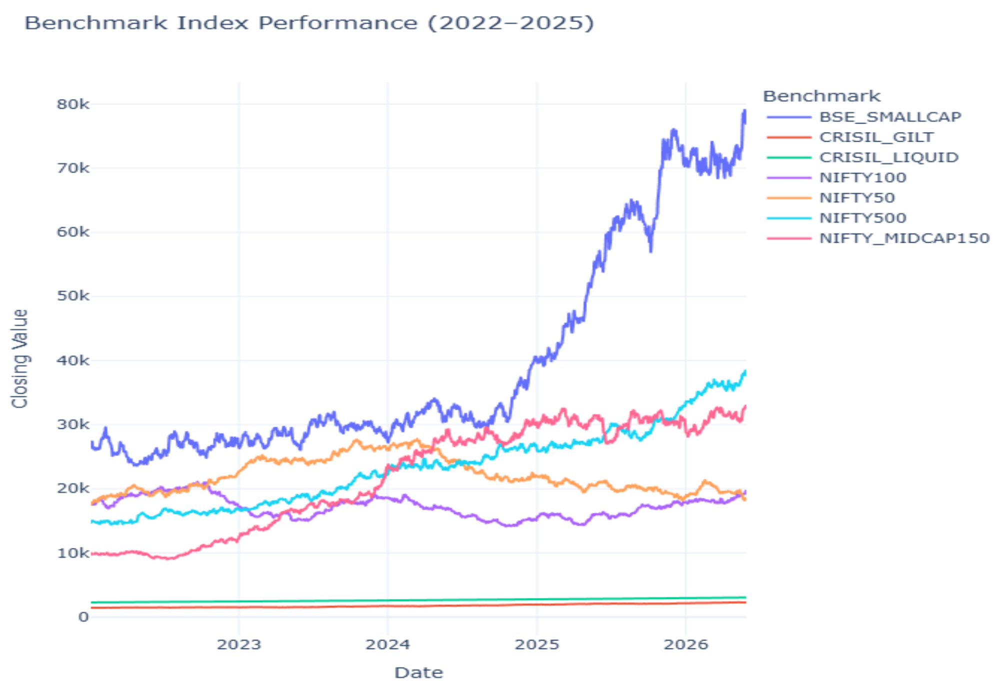


Figure 6.9: Benchmark Index Performance

Description

Benchmark indices provide a reference for evaluating mutual fund performance. Comparing fund returns with benchmark movements enables the calculation of Alpha, Beta, and Tracking Error in subsequent analytical stages.

Summary of EDA

The exploratory analysis revealed several important observations:

- Industry AUM and SIP inflows exhibited sustained growth over the analysis period.
- Retail investor participation increased steadily, as reflected by rising folio counts.
- Portfolio holdings demonstrated meaningful sector diversification.
- Mutual funds displayed varying degrees of correlation, indicating opportunities for portfolio diversification.
- Benchmark indices generally followed similar long-term market trends, providing a suitable basis for comparative performance analysis.

Performance Analytics

Performance analytics was carried out to evaluate the historical performance and risk characteristics of the selected mutual fund schemes. Multiple financial metrics were computed using historical NAV data to measure returns, volatility, downside risk, and benchmark-relative performance. These metrics provide a comprehensive assessment of fund quality beyond absolute returns and serve as the foundation for the advanced analytics and dashboard modules.

The analytical outputs were generated using Python and exported as structured CSV files, enabling seamless integration with the Power BI dashboard and subsequent analytical workflows.

7.1 Daily Returns

Daily returns were calculated using consecutive NAV observations to measure the percentage change in fund value from one trading day to the next. These returns formed the basis for the computation of all subsequent performance and risk metrics.

The daily return series captures short-term market fluctuations and provides the input required for volatility estimation, Sharpe Ratio, Sortino Ratio, Value at Risk (VaR), and other statistical analyses.

7.2 Compound Annual Growth Rate (CAGR)

Compound Annual Growth Rate (CAGR) was computed to measure the annualized growth of each mutual fund over the analysis period. Unlike simple average returns, CAGR represents the compounded rate of return while accounting for the actual number of trading days in the dataset.

Using CAGR enables fair comparison between schemes by expressing long-term performance as an annualized percentage.

7.3 Standard Deviation

Annualized standard deviation was calculated as the primary measure of fund volatility. It quantifies the variability of daily returns and provides an estimate of the investment risk associated with each scheme.

Funds exhibiting higher standard deviation generally experience larger price fluctuations, while lower values indicate relatively stable performance.

7.4 Sharpe Ratio

The Sharpe Ratio measures risk-adjusted return by comparing excess returns against total portfolio volatility. It indicates how efficiently a fund generates returns for each unit of risk undertaken.

Higher Sharpe Ratios generally represent superior risk-adjusted performance and were later used as one of the primary metrics within the recommendation system.

7.5 Sortino Ratio

Unlike the Sharpe Ratio, the Sortino Ratio considers only downside volatility by penalizing negative return deviations. This provides a more focused evaluation of downside investment risk while ignoring positive return fluctuations.

The Sortino Ratio is particularly useful for investors seeking consistent returns with limited downside exposure.

7.6 Alpha and Beta

Alpha and Beta were calculated by comparing mutual fund returns with their respective benchmark indices.

- **Alpha** measures excess return generated beyond benchmark expectations.
- **Beta** measures the sensitivity of a fund's returns relative to market movements.

Together, these metrics provide insight into both investment skill and market exposure.

7.7 Maximum Drawdown

Maximum Drawdown represents the largest peak-to-trough decline experienced by a fund during the observation period. It serves as an important measure of downside risk and helps investors understand the worst historical capital loss that could have been experienced.

Funds with smaller maximum drawdowns generally demonstrate greater resilience during adverse market conditions.

7.8 Tracking Error

Tracking Error was calculated by measuring the standard deviation of the difference between fund returns and benchmark returns. This metric quantifies how closely a fund follows its benchmark and is particularly important when evaluating index-oriented investment strategies.

Lower tracking error indicates closer benchmark replication, whereas higher values reflect greater deviation from benchmark performance.

7.9 Composite Fund Scorecard

A composite Fund Score was developed by combining multiple performance indicators including CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown. Each metric was normalized before being incorporated into the final scoring framework.

The scorecard enables comprehensive comparison of mutual funds by considering both return and risk characteristics instead of relying on a single performance metric. It also served as one of the primary datasets used within the interactive dashboard.

Summary

The performance analytics module generated a comprehensive set of financial indicators describing return, volatility, downside risk, benchmark performance, and overall fund quality. These metrics provide a robust quantitative foundation for evaluating mutual fund performance and support the advanced analytical techniques presented in the following section.

Metric	Purpose
Daily Returns	Percentage change in NAV
CAGR	Annualized growth rate
Standard Deviation	Volatility measurement
Sharpe Ratio	Risk-adjusted return
Sortino Ratio	Downside risk-adjusted return
Alpha	Excess return over benchmark
Beta	Market sensitivity
Maximum Drawdown	Largest historical loss
Tracking Error	Benchmark deviation
Fund Score	Overall composite ranking

Advanced Analytics

Beyond traditional performance evaluation, advanced analytical techniques were implemented to assess downside risk, investor behavior, portfolio diversification, and investment recommendations. These analyses provide deeper insights into mutual fund performance and investor activity, supporting more informed investment decisions.

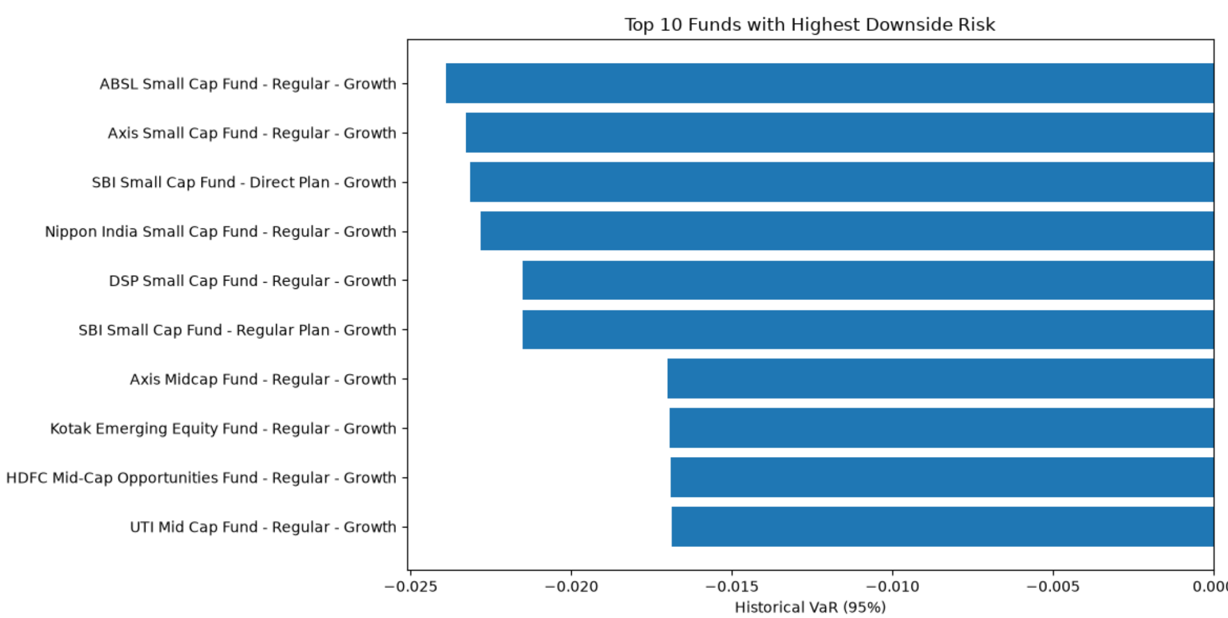
The outputs generated during this phase include risk reports, behavioral analytics, concentration analysis, and a rule-based recommendation system.

8.1 Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR)

Historical Value at Risk (VaR) at the 95% confidence level was calculated using the historical distribution of daily returns for each mutual fund scheme. VaR estimates the maximum expected loss under normal market conditions over a single trading day.

Conditional Value at Risk (CVaR), also known as Expected Shortfall, was computed as the average loss beyond the VaR threshold. Unlike VaR, CVaR measures the severity of losses during extreme market events and therefore provides a more comprehensive assessment of downside risk.

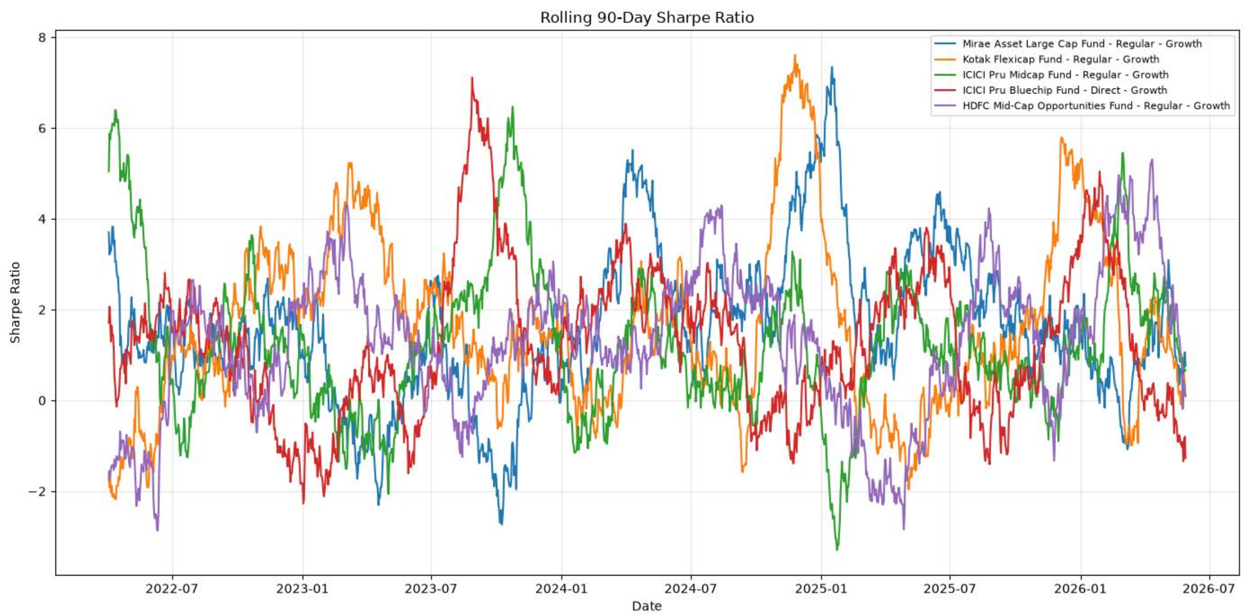
The generated VaR and CVaR reports allow comparison of downside risk across all analyzed schemes.



8.2 Rolling 90-Day Sharpe Ratio

A rolling 90-day Sharpe Ratio was calculated to analyze changes in risk-adjusted performance over time. Instead of representing each fund using a single Sharpe Ratio, the rolling analysis illustrates how performance evolved across different market conditions.

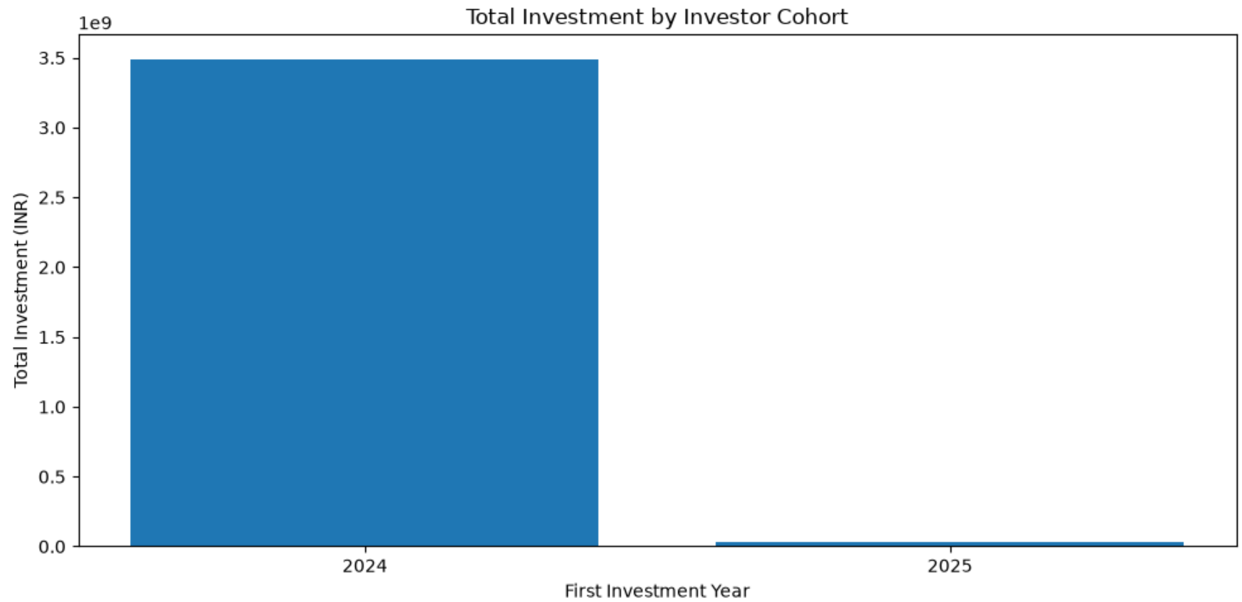
This time-series approach helps identify periods of improved or deteriorating fund performance and provides greater insight into investment consistency.



8.3 Investor Cohort Analysis

Investor cohort analysis was performed by grouping investors according to the year of their first recorded transaction. For each cohort, the average SIP amount, cumulative investment value, and most frequently selected mutual fund were determined.

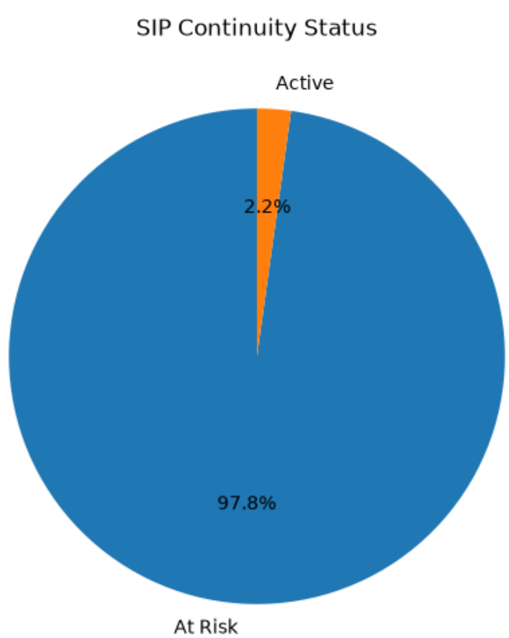
The analysis revealed behavioral differences between investor groups and highlighted how investment patterns varied across different entry periods.



8.4 SIP Continuity Analysis

To evaluate investment discipline, investors with six or more SIP transactions were analyzed based on the average interval between successive investments.

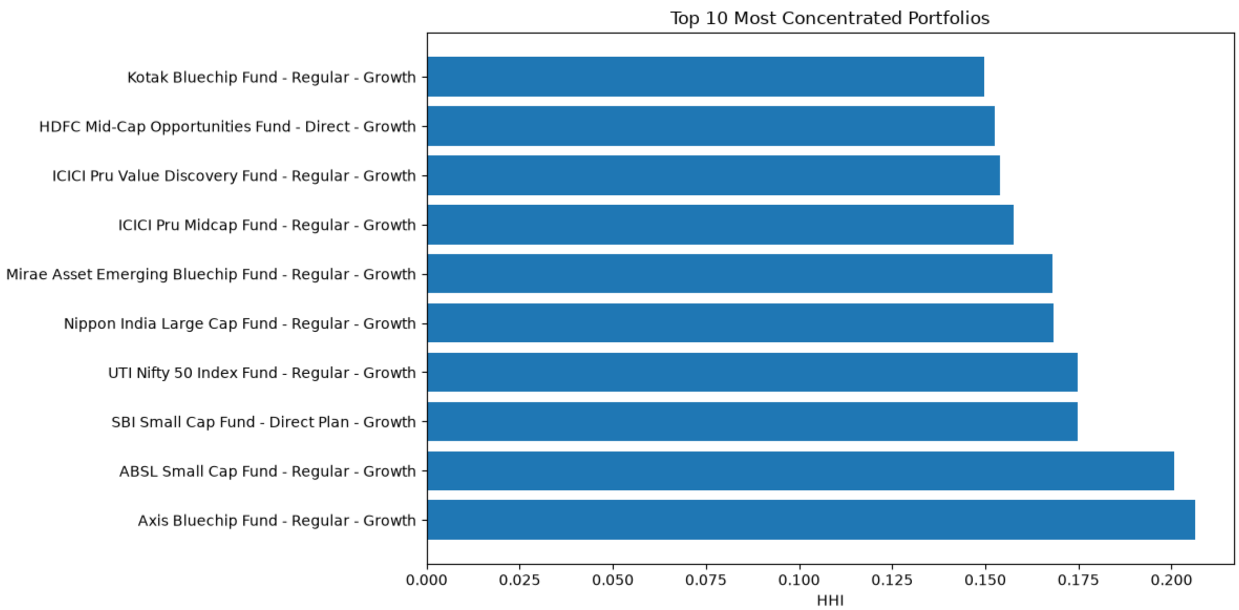
Investors whose average investment gap exceeded 35 days were classified as **"At Risk"**, indicating irregular contribution behavior. This analysis can assist fund managers in identifying investors who may require targeted engagement or retention strategies.



8.5 Portfolio Concentration Analysis

Portfolio concentration was measured using the Herfindahl-Hirschman Index (HHI), calculated from portfolio holding weights for each mutual fund.

Higher HHI values indicate greater concentration within fewer securities, whereas lower values represent more diversified portfolios. Comparing HHI values across funds provides additional insight into portfolio diversification and investment strategy.



8.6 Rule-Based Mutual Fund Recommendation System

A rule-based recommendation system was developed to assist investors in selecting suitable mutual funds based on their risk appetite.

The recommendation engine accepts one of three risk preferences:

- Low
- Moderate
- High

For each category, the system filters eligible mutual funds according to their risk grade and ranks them using the Sharpe Ratio. The three highest-ranked funds are then recommended along with their corresponding risk grade and composite fund score.

Although intentionally simple, the recommendation system demonstrates how quantitative performance metrics can be translated into practical decision-support tools for investors.

Key Insights

The advanced analytics module produced several important observations:

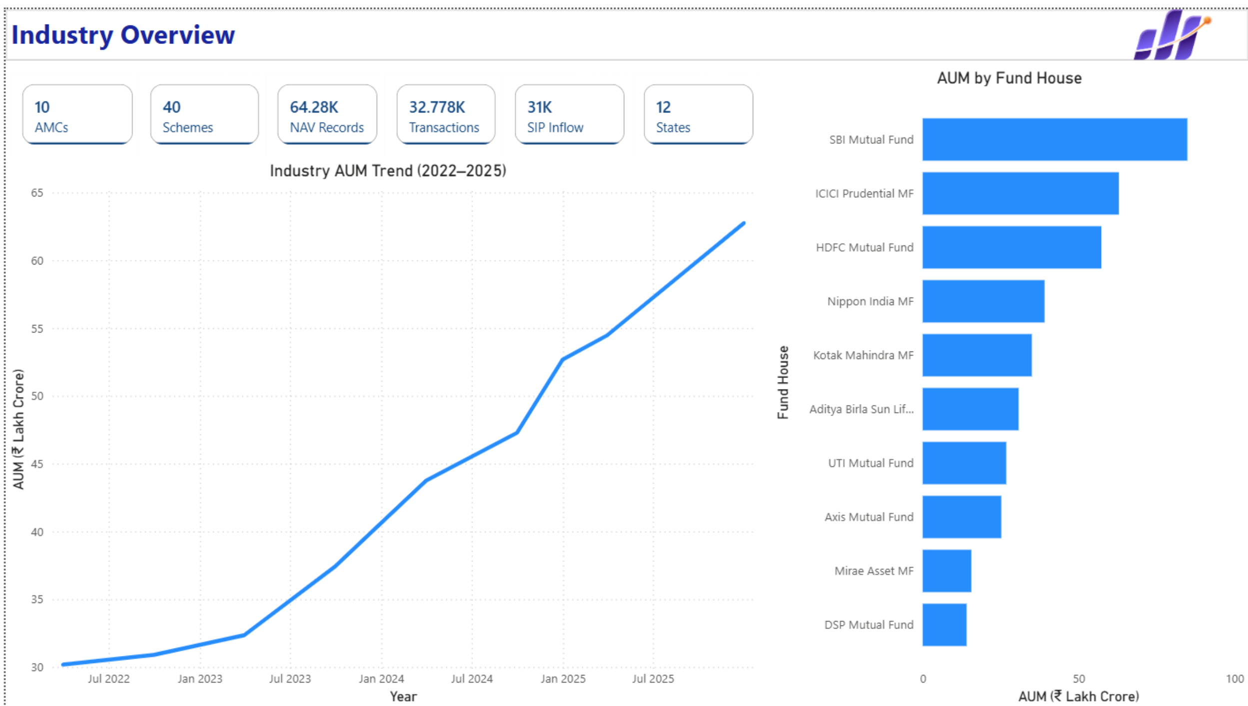
- Historical VaR and CVaR identified small-cap and mid-cap funds as exhibiting comparatively higher downside risk.
- Rolling Sharpe analysis showed that risk-adjusted performance varied significantly across different market conditions.
- Investor cohort analysis indicated that the 2024 cohort contributed the largest share of overall investments.
- SIP continuity analysis identified several investors with irregular contribution patterns, classifying them as **At Risk**.
- Portfolio concentration analysis revealed noticeable differences in diversification across mutual fund portfolios.
- The rule-based recommendation system successfully identified suitable funds for different investor risk profiles using risk-adjusted performance metrics.

Interactive Dashboard

An interactive dashboard was developed using Microsoft Power BI to transform the analytical outputs into an intuitive business intelligence solution. The dashboard integrates data from the processed datasets and analytical reports, enabling users to explore mutual fund performance, industry trends, investor behavior, and market insights through interactive visualizations.

Interactive slicers, drill-through functionality, and tooltips were incorporated to improve usability and support efficient data exploration.

9.1 Industry Overview



The Industry Overview page presents a high-level summary of the mutual fund industry using key performance indicators and trend visualizations.

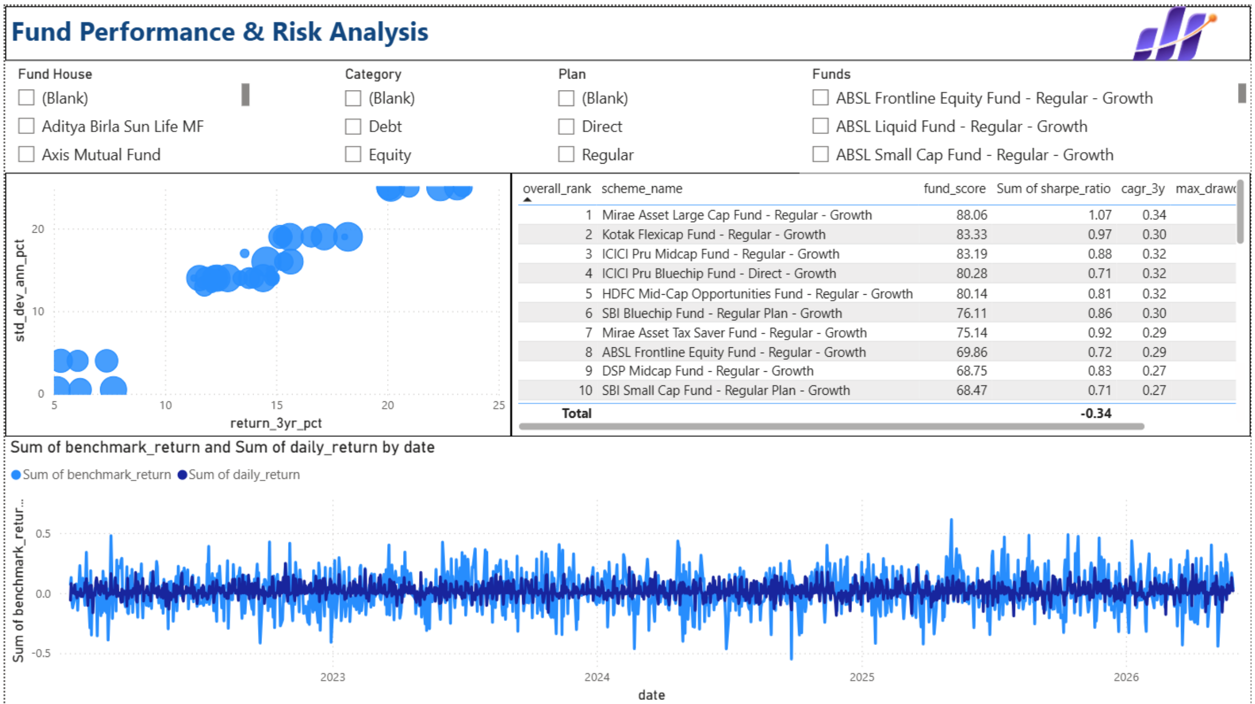
The dashboard includes:

- Total Assets Under Management (AUM)
- Monthly SIP Inflows
- Total Mutual Fund Folios
- Number of Active Schemes

- Industry AUM Trend
- AUM Distribution by Fund House

These visualizations provide a concise overview of industry growth and help users understand the overall market landscape.

9.2 Fund Performance



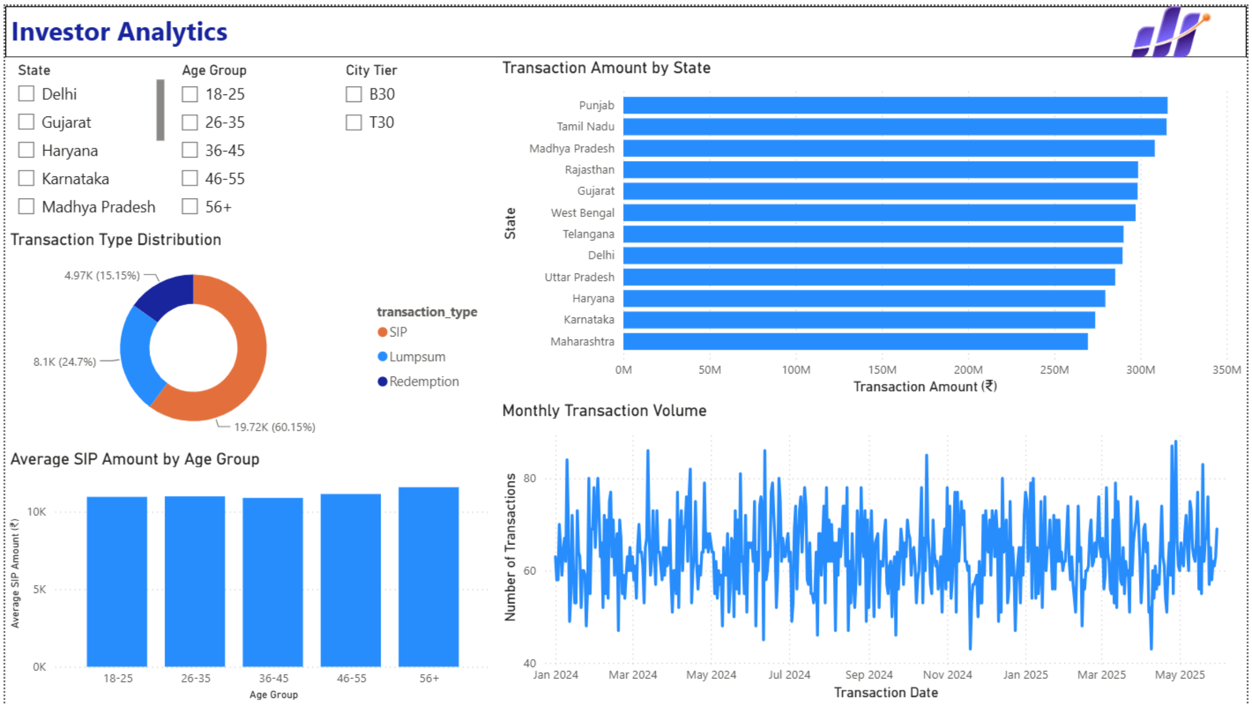
The Fund Performance page focuses on evaluating individual mutual fund schemes using multiple risk and return indicators.

Key visualizations include:

- Risk vs Return Scatter Plot
- Fund Performance Scorecard
- NAV versus Benchmark Comparison
- Interactive Filters for Fund House, Category, and Plan

This page enables investors to compare schemes based on both historical returns and risk-adjusted performance.

9.3 Investor Analytics



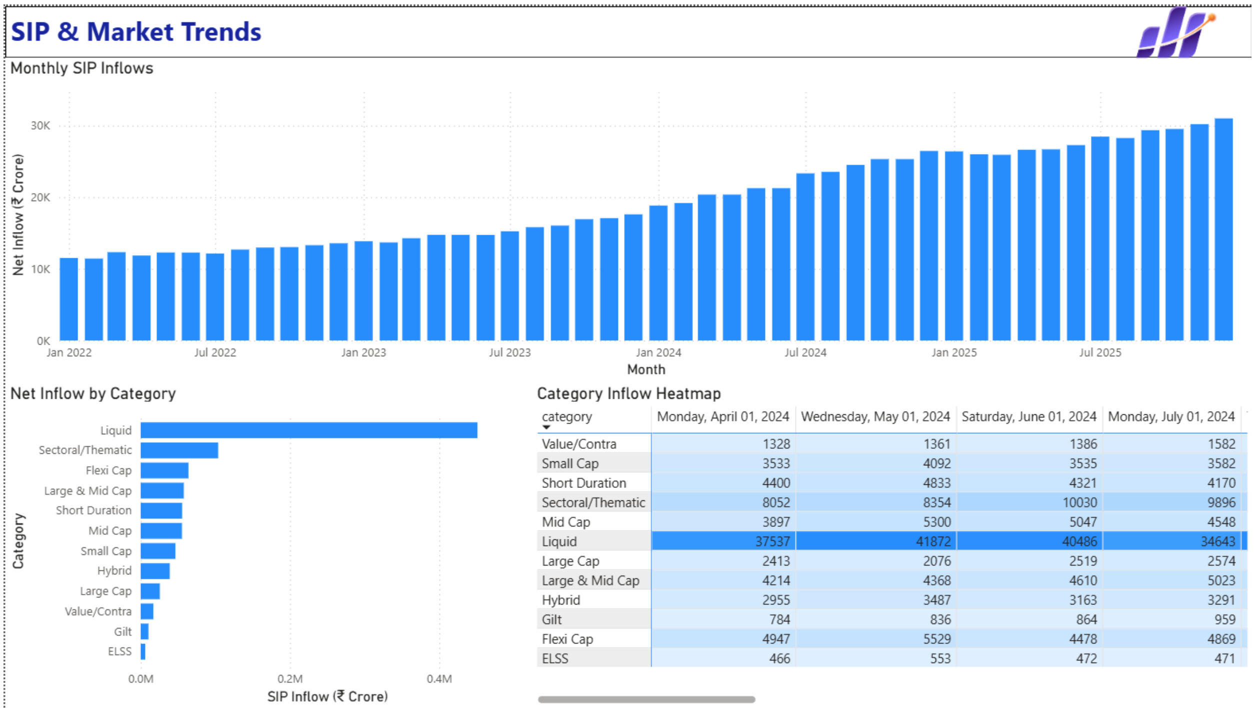
The Investor Analytics page explores investor demographics and transaction behavior.

The dashboard includes:

- Transaction Amount by State
- Transaction Type Distribution
- Average SIP Amount by Age Group
- Monthly Transaction Volume
- Interactive Filters for State, Age Group, and City Tier

These visualizations provide insight into investor participation patterns and regional investment behavior.

9.4 SIP & Market Trends



category	Monday, April 01, 2024	Wednesday, May 01, 2024	Saturday, June 01, 2024	Monday, July 01, 2024
Value/Contra	1328	1361	1386	1582
Small Cap	3533	4092	3535	3582
Short Duration	4400	4833	4321	4170
Sectoral/Thematic	8052	8354	10030	9896
Mid Cap	3897	5300	5047	4548
Liquid	37537	41872	40486	34643
Large Cap	2413	2076	2519	2574
Large & Mid Cap	4214	4368	4610	5023
Hybrid	2955	3487	3163	3291
Gilt	784	836	864	959
Flexi Cap	4947	5529	4478	4869
ELSS	466	553	472	471

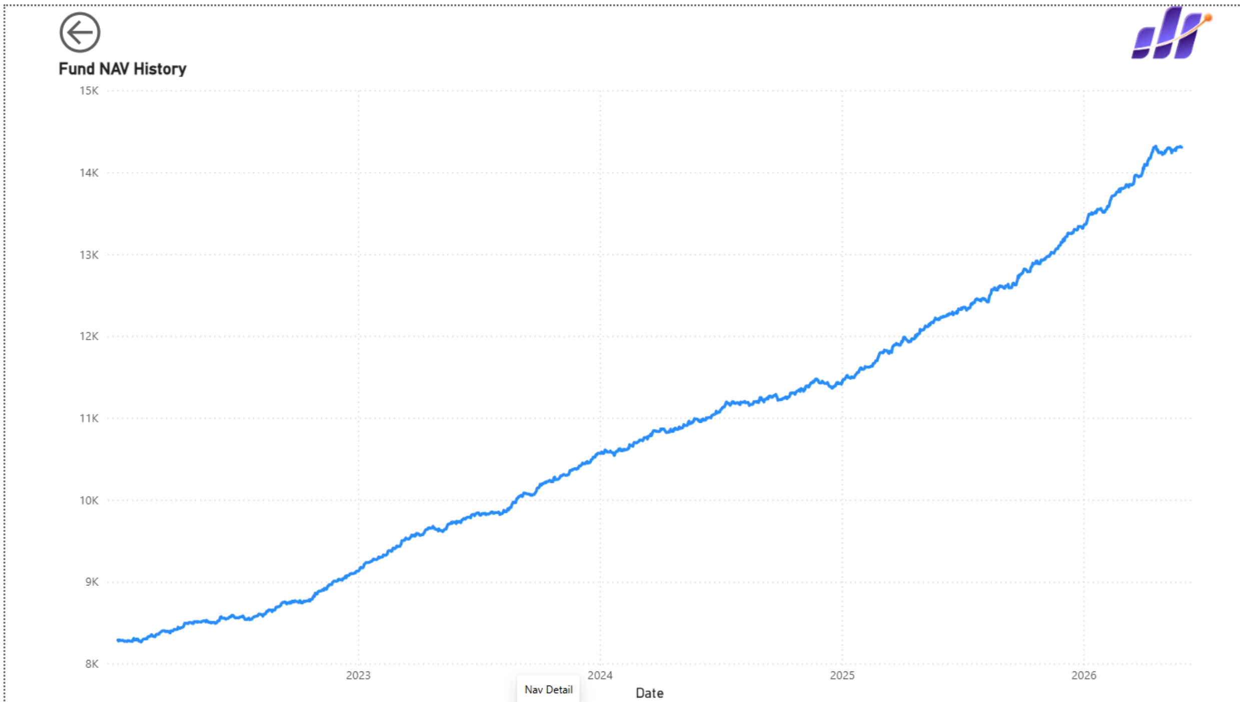
This page combines industry investment trends with market performance to analyze the relationship between investor activity and benchmark movements.

The dashboard includes:

- Monthly SIP Inflows
- Nifty 50 Trend
- Category-wise Inflows
- Top Investment Categories

The visualizations allow users to identify changes in investment behavior alongside broader market conditions.

9.5 NAV Detail (Drill-Through)



A drill-through page was developed to provide detailed analysis for individual mutual fund schemes.

Selecting a fund from the Fund Performance page automatically filters this page to display:

- Historical NAV Trend
- Scheme-specific performance
- Interactive navigation using the built-in Back button

This functionality enables users to move from summary-level insights to detailed scheme analysis without leaving the dashboard environment.

Dashboard Features

The Power BI dashboard incorporates several interactive features to enhance usability:

- Interactive slicers for filtering data by fund house, category, plan, state, and demographic attributes.
- Drill-through functionality for detailed fund-level analysis.
- Tooltips providing additional contextual information.

- Consistent Bluestock-themed design across all dashboard pages.
- Integration with the analytical datasets generated during the ETL and performance analytics stages.

Dashboard Summary

The interactive dashboard serves as the final presentation layer of the project, transforming processed financial data into meaningful visual insights. By combining industry statistics, performance analytics, investor behavior, and advanced analytical outputs within a single interface, the dashboard enables efficient exploration of mutual fund data and supports data-driven decision making.

Key Findings

The project generated several important insights through exploratory analysis, performance evaluation, advanced analytics, and interactive visualization. The major findings are summarized below:

- The mutual fund industry demonstrated sustained growth in Assets Under Management (AUM), SIP inflows, and investor folios throughout the analysis period, indicating increasing retail participation.
- Historical performance analysis showed that different mutual fund categories exhibited varying levels of return and risk, highlighting the importance of evaluating investments using multiple performance metrics rather than returns alone.
- Risk-adjusted performance metrics such as the Sharpe Ratio and Sortino Ratio enabled more balanced comparison between mutual funds by considering both returns and investment risk.
- Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) identified small-cap and mid-cap funds as having comparatively higher downside risk during adverse market conditions.
- Rolling 90-Day Sharpe Ratio analysis demonstrated that fund performance changes over time, emphasizing the importance of continuous monitoring rather than relying on a single historical value.
- Investor Cohort Analysis showed that the 2024 cohort contributed the largest share of total investments, reflecting stronger investment activity among earlier participants.
- SIP Continuity Analysis identified investors with irregular contribution patterns by measuring the average gap between successive SIP transactions, enabling identification of potentially inactive investors.
- Portfolio concentration analysis using the Herfindahl-Hirschman Index (HHI) highlighted differences in diversification across mutual fund portfolios.
- The rule-based recommendation system successfully recommended mutual funds according to investor risk appetite by combining risk grades with Sharpe Ratio rankings.
- The Power BI dashboard successfully consolidated financial metrics, investor analytics, and industry trends into an interactive platform that supports efficient data exploration and decision making.

Limitations and Future Scope

Although the developed platform provides a comprehensive mutual fund analytics solution, several limitations should be acknowledged.

- The analysis was performed using the datasets provided for the capstone project, which include synthetic investor transaction data and may not fully represent real-world market behavior.
- Live NAV data was incorporated only for selected schemes. Continuous real-time synchronization with external financial APIs was beyond the scope of this project.
- The recommendation system follows a rule-based approach using predefined risk categories and Sharpe Ratio rankings. It does not incorporate personalized financial objectives, investment horizons, or investor preferences.
- Portfolio optimization techniques and predictive modelling were not implemented, allowing opportunities for future enhancements.

Despite these limitations, the developed system successfully demonstrates an end-to-end workflow for mutual fund analytics and provides a strong foundation for future development. Several enhancements can be incorporated to further improve the platform and expand its analytical capabilities.

- Develop a fully automated ETL pipeline with scheduled data ingestion and periodic updates from live financial APIs.
- Integrate real-time market data to provide continuously updated dashboards and analytical reports.
- Extend the recommendation system using machine learning techniques that incorporate investor behavior, historical preferences, and personalized financial goals.
- Implement Modern Portfolio Theory and Markowitz Efficient Frontier optimization to recommend optimal investment portfolios.
- Develop a web-based analytics platform using Streamlit or similar frameworks to improve accessibility beyond Power BI.
- Incorporate predictive models and Monte Carlo simulations to estimate future portfolio performance under varying market scenarios.

These enhancements would further improve the practical applicability of the platform for investors, analysts, and financial institutions.

Conclusion

This project successfully developed a comprehensive Mutual Fund Analytics Platform that integrates data engineering, financial performance analysis, advanced risk analytics, and business intelligence into a unified workflow.

Beginning with raw financial datasets, an ETL pipeline was implemented to clean, transform, and organize the data for analysis. Exploratory Data Analysis provided insights into industry growth, investor behavior, and portfolio characteristics, while the performance analytics module computed key financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Tracking Error.

The advanced analytics module extended the platform by incorporating Historical VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration Analysis, and a rule-based mutual fund recommendation system. Finally, an interactive Power BI dashboard was developed to present these analytical results through intuitive visualizations and interactive exploration.

Overall, the project demonstrates how data engineering, quantitative finance, and business intelligence techniques can be combined to transform raw financial data into meaningful insights that support informed investment decisions. The modular design also provides a scalable foundation for future enhancements involving real-time analytics, portfolio optimization, and intelligent recommendation systems.